

# Academic Wasteland: a federated research commons for discovery, delegation and local control

## Author list open

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## Abstract

Using another laboratory's scientific services requires researchers to discover capabilities, translate requests, coordinate execution and respect restrictions on data use. Shared conventions could let agents perform this coordination while each laboratory retains control of its resources. During DBCLS BioHackathon 2026, we developed Academic Wasteland, a prototype research commons organised into independently configurable nodes called towns. The implementation combines authenticated message exchange, provider-owned semantic service descriptions, receiver-controlled authorisation, and versioned conformance profiles. FAIRhaven provides discovery and bounded service checks, while Camelot hosts reviewed credential issuance. Demonstrations cover permission-gated cohort comparison and a visitor's synthetic data processed on remote compute. A synthetic rare-disease workflow connects phenotype-based gene ranking at one town with local variant filtering at another and returns a report based on pinned public expert evidence. In the documented demonstration, FBN1 ranked third among 1,529 mouse gene profiles. Four of six synthetic variant calls passed filtering, and a specified combination of phenotype rank and variant score placed the selected FBN1 allele first. The workflow exposes its inter-town messages and keeps the complete variant file within the local analysis step. Separate executable certification scenarios exercise scoped trust, holder binding, delegation and revocation. These results establish a working integration of discovery, local policy and bounded scientific services. The present deployment uses a shared relay and co-hosted demonstration towns; evaluation across independent implementations, stronger host isolation and complete reproducibility packaging remain subsequent milestones.

**Working submission draft. The author list, affiliations, contribution statements and author order remain open.**

## Introduction

Scientific expertise, data and computational resources are distributed across laboratories. Consider a researcher investigating a possible genetic cause of a phenotype: a specialist service may maintain a useful prediction model, another institution may hold relevant variants under access restrictions, and a third group may provide an independent interpretation. Bringing these contributions together requires finding the services, establishing compatible inputs, obtaining permission, coordinating execution and understanding the evidence returned. As more scientific methods become accessible through agents, the ability to organise such collaborations becomes a research infrastructure problem.

A researcher can already coordinate these steps through email, web interfaces and scripts. Email can carry the question, data and answer, but the person must identify the recipient, translate each request, relay intermediate results and track the conditions under which they may be used. Bespoke scripts can automate an agreed collaboration, at the cost of encoding each service's interface and assumptions. We want a researcher to state an objective and let compatible services coordinate the routine steps under explicit local rules, while the researcher controls the objective, permitted actions and interpretation of the result. The practical motivation is to reduce the repeated integration work needed to use another laboratory's capabilities.

Local control is essential to this goal. A laboratory should be able to contribute a model or analysis service while retaining its choice of software, hardware and trusted collaborators. A data custodian should be able to permit a particular computation and release a limited result. A collaborator should be able to inspect who requested an analysis, which service performed it and which inputs and evidence support its output. Shared descriptions and linked research objects could also make small contributions easier to attribute and results easier to repeat or challenge. These are design objectives whose operational costs and benefits require evaluation.

The resulting vision is a research commons in which temporary collaborations can form around questions. Participation would consist of exposing a bounded capability under the provider's own conditions. This gives individual laboratories a reason to join before a complete ecosystem exists: they can offer an existing service and obtain complementary services through the same conventions. Common interfaces could reduce dependence on one agent framework or platform, while independent operation could preserve institutional responsibility for data and compute. The immediate question is which conventions are sufficient to make one useful collaboration work.

We investigated this question during [DBCLS BioHackathon 2026](#), held in Matsuyama, Japan, on 13–19 September 2026 (Database Division for Life Science, [2026](#)). A pre-hackathon proposal and semantic checker provided the starting point. During the event, we developed Academic Wasteland: communicating research nodes called towns, semantic service catalogues, scoped credential checks and executable demonstrations. This report describes the implemented connections between discovery, delegation and local admission decisions. Cohort comparison, remote execution and a phenotype-to-variant workflow test bounded instances of the intended collaboration; separate scenarios test certification failures. The software is available through the [Academic Wasteland organisation](#).

## Related work

### Scientific agents and reusable research methods

Agent Laboratory organises literature review, experimentation and report writing into a workflow of specialised language-model agents, with human feedback at each stage (Schmidgall et al., [2025](#)). Co-Scientist uses generation, critique, ranking and refinement agents to develop hypotheses from a scientist's objective and evaluates proposals in biomedical applications (Gottweis et al., [2026](#)). These systems show how agents can coordinate scientific reasoning within an orchestrated application. Wasteland addresses the deployment boundary around such capabilities: how separately operated services advertise themselves, receive requests and apply their own permissions. Either style of research assistant could act as a client or provider in that setting.

Paper2Agent is particularly close to the intended ecosystem. It converts papers and associated code into tested Model Context Protocol (MCP) tools and paper-specific agents, and demonstrates collaboration among multiple agents for gene prioritisation (Miao et al., [2026](#)). It therefore already addresses executable dissemination and cooperation among scientific methods. Wasteland concentrates on the operational conditions for using such methods across custodians: provider-owned discovery metadata, receiver-selected trust and control over intermediate disclosures. Connecting a Paper2Agent service would be a concrete interoperability test.

### Agent protocols and semantic service discovery

MCP standardises access to tools, resources and prompts, and specifies OAuth-based authorisation for protected HTTP services (Model Context Protocol contributors, [2025a](#), [2025b](#)). Agent2Agent (A2A) defines Agent Cards, messages, tasks and interaction patterns between agents. Its v1.0.0 specification requires authentication and authorisation checks and leaves the detailed policy to the server (Agent2Agent project contributors, [n.d.](#)). These protocols

provide relevant communication and security mechanisms. Wasteland makes application-level scientific requirements explicit, such as dataset-specific grants, semantic input types and checks before result release. The demonstrated deployment uses its own documented relay envelope; standards-based transport interoperability remains a separate conformance task.

Semantic discovery also has a substantial history in bioinformatics. Semantic Automated Discovery and Integration (SADI) describes services through Resource Description Framework (RDF) data and Web Ontology Language (OWL) input and output classes and supports their discovery and composition (Wilkinson et al., 2011). FAIRhaven follows this established motivation for machine-readable service descriptions, using a restricted JSON for Linked Data (JSON-LD) profile with Data Catalog Vocabulary (DCAT) and Semantic Science Integrated Ontology (SIO) terms. Its implemented semantic search retrieves candidate services, while installed operation adapters perform the demonstrated planning. General semantic workflow composition is a further development objective.

### Federated execution, permissions and provenance

The Global Alliance for Genomics and Health (GA4GH) Task Execution Service (TES) provides a common interface for batch tasks across computing environments, including execution near distributed data (Kanitz et al., 2024). Its Workflow Execution Service (WES) exposes workflow submission and management (Global Alliance for Genomics and Health, n.d.). These are natural execution backends for a research node. Wasteland's contribution lies in connecting the surrounding discovery, delegation and admission decisions to such local execution capabilities; a GA4GH adapter would require its own implementation and tests.

GA4GH Passports encode researcher attributes and access permissions as machine-readable visas for federated use (Voisin et al., 2021). Wasteland's credential model addresses related requirements, with separate claims and bindings to an agent holder, exact task and receiver. Passport translation should preserve issuer trust and the meaning of each permission. The Agent Identity Protocol (AIP) preprint also proposes invocation-bound tokens and attenuated delegation across MCP, A2A and HTTP (Prakash, 2026). This overlap makes cross-protocol authorisation a candidate for reuse and comparison. Wasteland's current credential encoding remains project-specific.

Workflow Run RO-Crate packages computational inputs, outputs, software and execution provenance, with implementations across several workflow systems (Leo et al., 2024). It provides a suitable foundation for the proposed packaging of a complete Wasteland interaction. At present, the prototype links messages and produces individual validation and interpretation reports. Together, these predecessors locate the contribution of this hackathon in an executable integration of established ideas: discovering a scientific capability, checking authority to use it, carrying out the allowed operation and preserving evidence of the exchange.

### Relationship to earlier BioHackathon projects

The BioHackathon series has addressed distributed scientific services since its early meetings. The BioHackathon 2008 report describes incompatible service interfaces and exchange formats, with BioMoby and workflow tools as integration mechanisms (Katayama et al., 2010). BioHackathon 2010 extended this agenda to linked data, identifiers, ontologies and semantic clients (Katayama et al., 2013). Wasteland continues that integration problem at the point where an agent chooses and requests an operation under another provider's policy.

The 2019 logic-programming group connected Prolog and SPARQL resources, extended Biolink representations and explored type inference (Mungall et al., 2020). This is a methodological predecessor for explicit machine-interpretable contracts. Wasteland uses a restricted message vocabulary and receiver-controlled ontology to decide task admissibility, alongside procedural checks for credentials and execution.

Two BioHackathon Europe 2023 submissions provide direct reuse opportunities. The WfExS/Workflow Run Crate report describes an ecosystem of workflow creators, data providers, registries, execution and deposition services (Fernández et al., 2024). The FAIR Digital Objects report connects RO-Crate, Bioschemas and Signposting for discoverable research objects (Soiland-Reyes et al., 2024). Wasteland adds an agent-facing request and permission exchange around these roles; consuming and producing their interoperable packages remains an implementation milestone. The Japan 2025 report *MCP server tools with RDF shapes* develops TogoMCP, schema context and rudof integration for natural-language access to RDF resources (Labra Gayo et al., 2025). Such a service could supply a town's knowledge-graph capability. Its schema context guides query construction; Wasteland's receiver contracts decide whether a submitted task is admissible.

## Position within the state of the art

Table 1 separates the established capability in each research line from the integration tested here. The comparison concerns responsibilities and evidence; comparative performance evaluation is future work.

**Table 1:** Established approaches and Wasteland's present scope.

| Research line                               | Established capability                                                  | Scope of the Wasteland prototype                                                                                            |
|---------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Agent Laboratory, Co-Scientist, Paper2Agent | Scientific reasoning, tool use and cooperating agents                   | Services operate behind separately configurable town policies; the demonstrated planner uses installed adapters.            |
| MCP, A2A and AIP                            | Tool and agent interfaces, authentication hooks and delegated authority | Scientific claims, task binding and release gates; current relay and credential profiles need cross-implementation testing. |
| SADI and BioHackathon RDF/MCP work          | Typed service discovery, composition and schema-guided queries          | Provider-owned descriptions plus bounded discovery; general semantic planning remains prospective.                          |
| GA4GH TES/WES and Passports                 | Portable execution and federated access information                     | Custodian and executor decisions joined in a working application; standards adapters remain prospective.                    |
| WfExS, RO-Crate and FAIR Digital Objects    | Reproducible execution, provenance packaging and discovery              | Linked message and decision traces; complete interoperable run packages remain prospective.                                 |

## Implementation

### Towns and message exchange

A town is an operational boundary containing services or agents, local resources and policies. A resident is an addressable service or agent within a town. This distinction lets a researcher contact a service without prescribing its internal model, workflow engine or programming language.

The hackathon deployment uses a shared HTTPS relay. Each town authenticates using its own bearer token. Messages contain a unique identifier, sender, recipient, timestamp, message kind and structured body. Replies reference the original request and reverse its endpoints. The relay checks the authenticated sender and rejects conflicting reuse of an identifier. Workers preserve replies before transmission so that interrupted delivery can retry the same message. This provides durable message handling; local handler side effects still require their own recovery strategy (Academic Wasteland contributors, 2026i).

The relay transports messages, while receiving towns select and execute handlers. Its credential establishes the sending town's transport identity. Agent qualifications and permissions are checked separately. This separation permits a small message client to participate while leaving scientific execution under the receiving operator's control.

### **FAIRhaven: discovering capabilities and inspecting evidence**

FAIRhaven applies the Findable, Accessible, Interoperable and Reusable (FAIR) principles to a practical question: which provider offers an operation with suitable scientific inputs and outputs, and what evidence accompanies that description? Providers publish their own service and dataset catalogues. Each entry identifies its publisher, operation, semantic types, formats, constraints, version, documentation and reuse conditions. The Python service harvests these catalogues over authenticated town messages, stores snapshots and revision history in SQLite, and exposes a web catalogue, JSON-LD and machine-readable search (Academic Wasteland contributors, 2026c). Another operator can run an index using the same implementation.

Registration fetches the requesting town's complete catalogue, validates publisher identity, identifier scope, locally controlled JSON-LD context and revision, then returns a persistent receipt. Malformed updates leave the previous snapshot intact; withdrawal and failed contact preserve an inspectable history. A pinned input/output type hierarchy supports semantic search. The catalogue separates provider declarations, metadata checks and observed demonstrations. Licence gaps remain visible, and the provider retains access decisions.

An hourly auditor contacts towns sequentially using reviewed, bounded probes. It can inspect catalogue replies, run synthetic phenotype examples and verify small public resource downloads. It labels unsupported operations or exhausted budgets as untested. Scheduling, reports and notification retries persist in SQLite. Providers receive private findings and metadata suggestions; the public interface exposes counts and timestamps. A successful probe establishes the specific response observed at that time (Academic Wasteland contributors, 2026d). This makes the difference between an advertised capability and a tested interaction visible to a prospective collaborator.

Concord complements FAIRhaven with versioned operating profiles and component-level conformance reports. Its initial holder-key and signed-status profiles contain nine and seventeen checks respectively. Reports identify the implementation, observer, profile and per-check outcomes. Receivers choose which profiles and observations they accept (Academic Wasteland contributors, 2026a).

### **Camelot: issuing scoped credentials after review**

Camelot is a town hosting a credential registrar and configurable demonstration issuers, including ethics and data-access authorities. It gives a requester a route from a missing permission to an application, a recorded human decision and a signed credential. Receivers independently choose which Camelot issuers, other institutions or local providers they trust for each claim (Academic Wasteland contributors, 2026e, 2026k).

The Python registrar maintains file-backed issuer descriptions, applications, credentials and revocation state. Public keys and credential material are separated from private signing keys. A requester supplies its holder identity and key, requested claim and purpose. The operator reviews the application through a command-line interface or protected local cockpit, approves or

denies it, and records the decision. Review evidence stays in private audit files. Approval issues an Ed25519-signed credential; receivers obtain its status through configured registrar endpoints. Public application submission and status retrieval remain separate from the operator's approval interface (Academic Wasteland contributors, 2026l).

Camelot's purpose is to make the source, scope and lifecycle of an authority claim explicit. The receiving town still checks the holder, required claim types, task and audience bindings, expiry and status. Demonstration ethics issuers exercise this mechanism with synthetic or public data. Institutional adoption would require recognised review procedures, accountable issuers and appropriately isolated services.

## Semantic contracts and receiver-controlled authorisation

The Research Commons prototype validates messages in layers. JSON Schema checks serialised structure. Shapes Constraint Language (SHACL) checks graph constraints. An OWL semantic gate tests consistency and entailment against a trusted, versioned ontology bundle using the Kobayashi-MaRust reasoner. Sender assertions are restricted to permitted assertions about individuals; the receiver controls the ontology and its own additional facts. The checker distinguishes entailed, unknown, invalid and indeterminate outcomes (Academic Wasteland contributors, 2026j).

Protected computation requires additional procedural and cryptographic checks. Credentials and signed presentations bind an agent to its key, the exact task and the receiving town. Each receiver specifies which issuer may attest which claim within which dataset and operation scope. Qualification, ethics approval, dataset permission, compute permission and human delegation are distinct claims. A policy may require several claim types together while accepting alternative trusted issuers for a particular type (Academic Wasteland contributors, 2026e).

Delegated issuer authority is bounded by claim type, scope, dataset and permitted delegation depth. Status is rechecked before execution and before releasing results. A Keycloak adapter maps authenticated, introspected issuer, subject and group information to configured qualifications. The credential signature encoding is a project-specific profile. Credential issuance therefore remains a substantive decision about evidence, and interoperability with other credential implementations requires additional profile agreement.

## Use cases and demonstrations

The demonstrations test distinct boundaries: service discovery, dataset permission, access to remote compute, restricted disclosure and issuer choice. Results below come from dated captured runs and implementation documentation. Supplementary demonstration notes provide replay instructions, screenshots and the evidence associated with each case.

### Comparing cohorts under separate permissions

A researcher requests allele counts for the same genomic region from two logical cohorts in the public JaSaPaGe VCF. Ubar manages a nine-sample Saudi view and Yamatai a ten-sample Japanese view. The local stage withholds Ubar's dataset permission initially. An operator approves the exact region and aggregate output, issuing a signed grant; two local workers then run `bcftools` and recheck credentials before release. The captured rehearsal returned 510 shared callable biallelic sites in GRCh38 chr6:29940000–29950000. Attempting individual-genotype export with the aggregate approval was refused before an export process started (Academic Wasteland contributors, 2026m).

This case demonstrates permission-dependent execution and output scope over real public-source data. Cohorts are selected through sample manifests, and the displayed counts describe these small sample sets. The stage uses local workers and isolated demonstration authorities. A

separate custody-adapter verification documents Yamatai executing Ubar's approved cohort task on DDBJ Slurm, with unsigned requests refused and successful-task replay returning the stored result (Academic Wasteland contributors, 2026b). That adapter uses short-lived custodian grants for public-source cohorts; its policy is distinct from the native controlled-computation certification path.

## A visitor obtains remote compute

A second case starts with a visitor who has data but needs another town's execution capability. The visitor grants dataset permission, Yamatai supplies compute permission, and an isolated Camelot ethics issuer supplies the remaining required claim. Before that approval, the workflow waits without transferring the file or submitting a job. The approval binds the exact input fingerprint, task and destination (Academic Wasteland contributors, 2026m).

The default input is a synthetic three-site, two-sample VCF. After approval, Yamatai transfers it and submits a bounded Slurm job through DDBJ. The documented rehearsals completed jobs 20604587 and 20604631; the browser proof verified alternate/called allele counts of 1/4, 3/4 and 1/4. Credentials and the input binding are checked before transfer and before release, and the workflow returns aggregate output while cleaning up individual-level staged files (Academic Wasteland contributors, 2026n). This demonstrates cooperation among a data holder, an execution provider and an accepted authority. The isolated approval is a test credential; production use requires the institution's actual governance process.

## Phenotype-guided analysis with local variant filtering

We used a synthetic Marfan teaching case to test whether a town could obtain phenotype-based gene ranks, combine them with a locally held variant file and request interpretation of the selected allele. The input phenotypes include ectopia lentis, arachnodactyly and aortic root aneurysm, supplied as labels paired with Human Phenotype Ontology identifiers. Owner-controlled resource metadata specifies permitted compute locations and recipients, including the prohibition on exporting the complete variant file (Academic Wasteland contributors, 2026h).

A contact service discovers suitable advertised operations and returns a bounded plan. Yamatai sends phenotype information to Ubar, where the installed INDIGENA model ranks mouse gene profiles and returns human orthologue annotations. Yamatai then opens its own synthetic Variant Call Format (VCF) file and filters its six heterozygous calls. Retention requires PASS status, read depth of at least 10, genotype quality of at least 20, allele frequency at most 0.001 and membership in the returned gene list. Four calls pass.

For each retained call, the demonstrator computes

$$s = \frac{0.45}{1 + \log_2(r)} + 0.55v,$$

where  $r$  is the phenotype-based gene rank and  $v$  is the REVEL variant score. This explicit heuristic combines the two evidence sources. The selected allele is evaluated after ranking; its identity is excluded from the inputs to the rankers.

The [captured live run](#) reports FBN1 at rank three among 1,529 mouse gene profiles. The FBN1 allele ranks second by REVEL alone and first under the combined heuristic. Table 2 shows how the phenotype rank changes the order of the leading two retained candidates (Academic Wasteland contributors, 2026g, 2026h).

**Table 2:** Comparison of the selected FBN1 allele and the synthetic ADAMTSL4 decoy in the teaching case. Combined scores are rounded.

| Candidate      | Phenotype gene rank | REVEL score | Combined score | Combined rank |
|----------------|---------------------|-------------|----------------|---------------|
| FBN1           | 3                   | 0.821       | 0.6256         | 1             |
| ADAMTSL4 decoy | 5                   | 0.860       | 0.6085         | 2             |

The selected allele is GRCh38 15:48421680 T>C, FBN1 NM\_000138.5:c.7577A>G (p.Asn2526Ser). Yamatai sends that allele and resolved phenotype identifiers to Themis, an interpretation service at Ubar. The complete VCF, genotypes and candidate shortlist stay within Yamatai's local analysis step. The service generates a PDF, which Yamatai checks against the returned digest. Inspectable request and response identifiers connect planning, analysis and report retrieval.

Themis uses pinned public expert evidence for this one allele. It reproduces the ClinGen FBN1 expert assessment using PS4, PM2\_Supporting, PP2 and PP3, yielding the classification "Likely pathogenic" under the specified American College of Medical Genetics and Genomics/Association for Molecular Pathology rule. The implementation documents the dated assessment and FBN1 specification used. Other alleles receive an explicit unclassified result. The synthetic case contributes no additional proband evidence; the classification concerns the curated variant–disease association (Academic Wasteland contributors, 2026h).

This demonstration establishes the operation sequence and the information exchanged at each boundary. The case and decoy were constructed to illustrate combined ranking. The current towns share a host and user account, so the demonstrated boundary is enforced at the protocol and application layers. The fixture is public synthetic data. Selected alleles and phenotype identifiers themselves remain disclosures that a resource owner must permit.

## Alternative issuers, impersonation and revocation

Separate executable scenarios test how different receivers apply scoped trust. The six scenarios cover local issuer choice, a shared independent issuer, claims supplied by separate authorities, bounded accreditation chains, holder impersonation and human delegation, and unavailable, expired or revoked certification. The captured scenarios show Lisan accepting its local issuer while refusing Camelot-only qualifications, two towns accepting an independent board, and a delegated issuer being limited to its authorised scope. Each scenario includes accepted and rejected paths. Controlled-computation tests also exercise revocation during toy VCF processing and refusal to release the computed result (Academic Wasteland contributors, 2026e).

A recorded Keycloak v26.7.3 run additionally accepted a configured service-account qualification, refused holder impersonation and refused a revoked token on fresh introspection (Academic Wasteland contributors, 2026f). These checks make the admission decision inspectable: a qualification satisfies one requirement, while dataset, compute and any required ethics or human permissions must also be satisfied. Strict certification is configurable on the native controlled-computation path. Public contact services and simpler starter workflows retain their own documented policies. The phenotype demonstration and strict certification scenarios therefore exercise complementary components of the prototype.

## Discussion

The hackathon produced a working connection between service descriptions, messages, local scientific operations and explicit receiver policy. The biological example makes the value of

this connection concrete: one town contributes phenotype-based ranking, another combines it with local variant data, and an interpretation service returns a sourced report. The human supplies the objective, and the installed adapters coordinate a bounded workflow with visible requests and results.

The implementation also clarifies three distinct forms of evidence. A relay token authenticates transport. A credential states a scoped claim from an issuer trusted by the receiver. A conformance report describes observations against a particular profile and implementation. Keeping these meanings separate helps an operator determine which evidence supports a specific decision.

Several boundaries define the next evaluation. The current relay is shared infrastructure, and the demonstration towns are co-hosted. Deployments under independent operators and administrative accounts would test operational federation and stronger isolation. Independent clients written from the protocol specification would test implementation interoperability. The constructed variant example establishes workflow behaviour; a separately designed dataset and evaluation protocol would support estimates of ranking performance. Themis currently reproduces one pinned expert assessment, making broader interpretation coverage a distinct development task.

The pre-hackathon proposal also envisaged independent replication, challenges to scientific claims and complete research-object packaging. The present report covers message linkage, validation reports, catalogue observations and a generated interpretation report. A subsequent milestone is to package an entire interaction as a reproducible RO-Crate and have an independently operated service repeat or challenge a result. Model checkpoints, ontology versions, external evidence and resource licences must be retained alongside the execution description for this purpose.

## Future work

The next interoperability test should connect a separately developed client and independently administered towns using a pinned profile. A2A transport, MCP service and GA4GH execution adapters should preserve task identities, permissions and release decisions across their boundaries. Tests should include incompatible semantic types, changed catalogue versions, unavailable status services and interrupted execution. Separate operating-system accounts and restricted execution services would extend the current application-level custody boundary.

A second milestone is a complete Workflow Run RO-Crate containing inputs or permitted references, workflow and model versions, signed decisions, execution observations and released outputs. An independent town should repeat the analysis and publish agreement or disagreement linked to that package. This would connect the prototype directly to the earlier WfExS and FAIR Digital Object work.

The practical benefit also needs measurement. A study should compare manual coordination, bespoke scripts and the town workflow on time to connect a new service, operator interventions, task completion, policy violations, compute cost and recovery from failure. FAIRhaven's probes need calibrated coverage and freshness measures. The phenotype workflow needs a held-out benchmark and broader interpretation evidence. Governance work should define issuer review responsibilities, revocation and appeal, software maintenance, attribution and incentives for providers to remain available.

## Conclusions

Academic Wasteland connects service discovery, reviewed credentials and receiver-controlled execution in a research commons prototype. The demonstrations show how these components support cohort analysis, remote compute and selective disclosure during variant prioritisation.

Existing scientific-agent systems and BioHackathon infrastructure provide reusable foundations; the next step is to test these connections across independent implementations and package their results for reproducible reuse.

## Software and data availability

The [Research Commons repository](#) contains the initial proposal, schemas, ontology contracts and semantic checker (Academic Wasteland contributors, 2026j). The [pangenome-town repository](#) contains the research services, certification implementation, synthetic variant fixture, evidence package and captured demonstrations (Academic Wasteland contributors, 2026e, 2026h). The [starter-pack repository](#) contains the relay client, worker, FAIRhaven and Concord implementations (Academic Wasteland contributors, 2026a, 2026c, 2026i). Versioned documentation links in the bibliography identify the source snapshots used for this draft.

The [recorded demonstration replay](#), [live cohort demonstration](#), [live visitor demonstration](#), [interactive phenotype demonstration](#), [FAIRhaven catalogue](#) and [Concord handbook](#) provide interfaces to the deployed prototype. Reproduction instructions and environmental requirements are recorded with the demonstration, including the installed INDIGENA checkpoint and running town bridges. Upstream data, model and evidence rights remain attached to their respective resources.

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